

Computer Networking Demo in 'C'

MANUJ ARORA

A computer network is a collection of independent computers that communicate with one another over a shared medium. It is the networking that enables people on a network to share files and peripherals such as modems, printers, back-up drives and CD-ROM drives. Moreover, with the use of services available from phone companies, people can send e-mails and access numerous services available over the Internet.

Here is a program that demonstrates the working of a computer network. To help you understand the concept better, let us first describe the different types of network classifications, topologies and Open Systems Interconnection (OSI) model.

Types of computer networks

Based on architecture. Networks can be divided into two categories based on architecture: Peer-to-peer and client-server.

Peer-to-peer. In a peer-to-peer (P2P) network, all the computers share equal responsibility for data processing (refer Fig. 1). Peers are equally privileged, equipotent participants in an application. They form a peer-to-peer network of nodes. There is no dedicated server or hierarchy among the computers.

Peers make a portion of their resources such as processing power, disk storage and network bandwidth directly available to other network participants without the need for central coordination by servers or stable hosts. Peer networks are often situated physically close to each other, and used typically in homes, small businesses or schools.

Client-server. Client-server network model comprises a centralised host (server) and multiple client computers (refer Fig. 2). In this network, all the

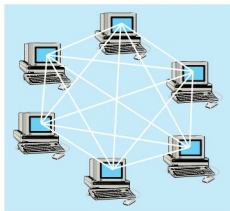


Fig. 1: Peer-to-peer network



Fig. 2: Clients connected to a host server

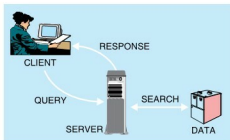


Fig. 3: Working of client-server model

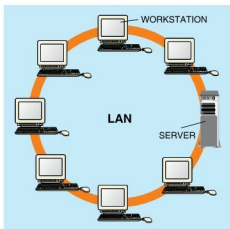


Fig. 4: Local-area network



information and files are centralised on the server and any client computer can access any information at any time, irrespective of the source computer's power status. Due to centralisation of data, information and resources, client-server networking provides centralised backup facilities, Internet facilities and the assurance of Internet monitoring.

The client initiates a transaction by sending a request (query) to the server. The server, which is always in a state of readiness to accept requests, responds by providing a service or resource to the client (refer Fig. 3).

Based on physical scope. There are many different types under this category:

Local-area network (LAN). LANs are networks usually confined to a geographic area such as a building, home, school or college campus. This network offers very high data transfer rates. LAN connection is shown in Fig. 4.

Metropolitan area network (MAN). This network spans over a metropolitan city area.

Wide-area network (WAN). WAN combines multiple LANs such that the network spans over a very large geographical area such as a country or even continent. This is accomplished

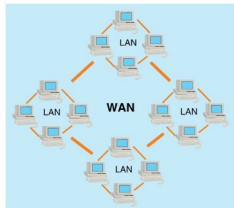


Fig. 5: WAN connection